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Serum 25-hydroxyvitamin D, vitamin D binding protein, and risk of colorectal cancer in the Prostate, Lung, Colorectal, and Ovarian Cancer Screening Trial

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Abstract

The potential role of vitamin D in cancer prevention has generated substantial interest, and laboratory experiments indicate several anti-cancer properties for vitamin D compounds. Prospective studies of circulating 25-hydroxyvitamin D [25(OH)D], the accepted biomarker of vitamin D status, suggest an inverse association with colorectal cancer risk, but with some inconsistencies. Furthermore, the direct or indirect impact of the key transport protein, vitamin D binding protein (DBP), has not been examined. We conducted a prospective study of serum 25(OH)D and DBP concentrations and colorectal cancer risk in the Prostate, Lung, Colorectal, and Ovarian Cancer Screening Trial, based on 476 colorectal cancer cases and 476 controls, matched on age, sex, race, and date of serum collection. All subjects underwent sigmoidoscopic screening at baseline and once during follow-up. Conditional logistic regression estimated odds ratios (ORs) and 95% confidence intervals (CIs). Circulating 25(OH)D was inversely associated with colorectal cancer (OR=0.60, 95% CI 0.38-0.94 for highest versus lowest quintile, p-trend 0.01). Adjusting for recognized colorectal cancer risk factors and accounting for seasonal vitamin D variation did not alter the findings. Neither circulating DBP nor the 25(OH)D:DBP molar ratio, a proxy for free circulating 25(OH)D, was associated with risk (OR=0.82, 95% CI 0.54-1.26, and OR=0.79, 95% CI 0.52-1.21, respectively), and DBP did not modify the 25(OH)D association. The current study eliminated confounding by colorectal cancer screening behavior, and supports an association between higher vitamin D status and substantially lower colorectal cancer risk, but does not indicate a direct or modifying role for DBP.

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Keywords

25-hydroxyvitamin D; vitamin D binding protein; colorectal cancer; serum biomarkers; prospective study

While it is known that vitamin D is critical for bone health,¹ laboratory evidence indicates that vitamin D also has many anti-cancer properties, such as reduction of inflammation, inhibition of cellular proliferation and angiogenesis, and promotion of cellular differentiation and apoptosis.^{2,3} Vitamin D is synthesized in the skin when exposed to ultraviolet B radiation (UVB)² or consumed from foods with naturally-occurring vitamin D, from foods fortified with vitamin D, or from dietary supplements. It is then hydroxylated in the liver to form 25-hydroxyvitamin D [25(OH)D], the accepted biomarker of vitamin D status.⁴ This 25(OH)D is further hydroxylated in the kidney and other organs to its active hormonal form, 1,25-dihydroxyvitamin D [1,25(OH)₂D], by the 1 α -hydroxylase enzyme, CYP27B1.³

Both circulating 25(OH)D and 1,25(OH)₂D are bound (about 88% and 85%, respectively) to vitamin D binding protein (DBP, also known as GC-globulin), the key transport protein, with small amounts bound to albumin, and very little of either vitamin D form in a free state.^{5,6} DBP also serves as a cochemotactic factor and an actin scavenger,^{5,6} and a deglycosylated form of DBP, known as the DBP-macrophage activating factor, has anti-carcinogenic properties operating via angiogenic, apoptotic, and immune pathways⁶⁻⁸ and has been shown to inhibit tumor growth in vivo and in vitro for several organ sites, including the large bowel.⁵⁻⁷ We have previously shown in cohort studies that circulating DBP is inversely associated with pancreatic and renal cancer risk,^{9,10} and that it modifies the associations between 25(OH)D and risk of pancreatic,⁹ prostate,¹¹ and bladder cancers.¹²

Meta-analyses of prospective epidemiologic studies indicate that circulating 25(OH)D is inversely associated with colorectal cancer risk.¹³⁻¹⁵ However, risk estimates from only a few investigations were statistically significant¹⁶⁻¹⁹ and one study found statistically significantly elevated risk for higher status.²⁰ In addition, several studies found inconsistent results for colon versus rectal cancer¹⁷⁻²² and few reported on associations by sex.^{18,21,22} Importantly, the impact of DBP, either as a modifier of the 25(OH)D association or as a direct determinant of risk, has not been examined for this cancer. In the current report, we describe a large study of circulating 25(OH)D and DBP, using prospectively stored serum samples, and subsequent risk of colorectal cancer in men and women in the United States.

MATERIALS AND METHODS

Study population

Participants in this nested case-control study were drawn from the Prostate, Lung, Colorectal, and Ovarian Cancer Screening Trial (PLCO) cohort, which was designed to investigate whether cancer screening tests reduce the mortality from these cancers. The community-based trial included 10 centers in the United States (Birmingham, AL; Denver, CO; Detroit, MI; Honolulu, HI; Marshfield, WI; Minneapolis, MN; Pittsburgh, PA; Salt Lake City, UT; St Louis, MO; and Washington, DC) which recruited participants, ages

55-74 years, between 1993 and 2001.²³ Individuals with prior cancer of the colon, rectum, prostate, lung, or ovary were ineligible. Participants were also excluded if their colon had been surgically removed, or if they had a colonoscopy, sigmoidoscopy, or barium enema in the three years prior to randomization (the latter only for participants randomized beginning in 1995).²⁴ Men (n=38,340) and women (n=39,105), randomized to the screening arm of the trial, were offered colorectal cancer screening by flexible sigmoidoscopy at the start of the trial, and again either three or five years later. Participants with an abnormal finding (ie, a polyp or mass) at either baseline or follow-up were referred to their medical-care providers for follow-up, but were not excluded from the study. The loss to follow-up for the Trial was 0.1%.²⁴ The PLCO Trial was approved by the institutional review boards of the U.S. National Cancer Institute and each of the 10 participating screening centers, the latter of which obtained written informed consent from each participant.²³

Data collection

A self-administered questionnaire was used to collect information at baseline on sociodemographic characteristics and risk factors such as height, weight, smoking history, family history of cancer, personal medical history (including history of colorectal polyps), and recent history of cancer screening exams. A 137-item food frequency questionnaire was used to gather information on usual dietary intake, including alcohol consumption and physical activity during the year prior to study enrollment.²³

Case identification and control selection

We included all colorectal cancer cases in the screening arm of the trial without a history of cancer, other than non-melanoma skin cancer, prior to their colorectal cancer diagnosis, who had a baseline serum sample collected at least one month prior to diagnosis, and who were diagnosed by July 31, 2011. Cancers were identified annually by self-report and confirmed by medical record abstraction. All cases with a history of colon diseases (ie, polyps, diverticulitis, ulcerative colitis, crohn's disease, and familial polyposis) prior to randomization were excluded. Cases (n=476) were defined as colon (C180, C182-C189 and C199; n=421) and rectal (C209 and C218; n=55) cancers based on the International Classification of Diseases for Oncology, 2nd edition (ICD-O-2). Cancers of the appendix (C181) were excluded. Proximal colon cancer (n=300) was defined as ICD-O-2 codes C180, and C182-C185, while distal colon cancer (n=119) was defined as ICD-O-2 codes C186-C188 and C199. Two cases could not be categorized as proximal or distal and were therefore excluded from these subanalyses. Stage coding was based on the American Joint Committee on Cancer 5th edition.²⁵

Controls (n=476) were also drawn from the screening arm of the trial and, therefore, underwent the same screening protocol as the cases. Potential controls (n=47,671) were alive and free of cancer or colon diseases, including polyps, up to the time of the matched case's diagnosis date and had a baseline serum sample collected at least one month prior to this date. One control was selected randomly for each case and matched on age at study entry (+/- 1 year), sex, race [non-Hispanic white, non-Hispanic black, other (consisting of Hispanic, Asian, Pacific Islander, and American Indian)], and date of serum collection (+/- 30 days, but expanded to +/- 60 days for two subjects).

Laboratory analyses

Serum samples were collected at baseline, shipped to a central biorepository, and stored at -70°C .²³ Circulating 25(OH)D was measured at Heartland Assays, Inc. (Ames, IA) with a direct, competitive chemiluminescence immunoassay (DiaSorin Liaison 25(OH)D TOTAL assay).²⁶ DBP was measured using a monoclonal Quantikine Human Vitamin D Binding Protein Immunoassay Kit (catalog number DVDBP0; R&D Systems, Inc.) at the Frederick National Laboratory for Cancer Research (Frederick, MD). Matched case-control pairs were placed in adjacent positions in the same batch and each batch also contained masked quality control materials comprising ~10% of the total samples. Half of the masked samples were from a PLCO serum pool and half were National Institute of Standards and Technology standard reference materials (SRM 968) representing concentrations of 18, 32, and 50 nmol/L of 25(OH)D. Interbatch, intrabatch, and total coefficients of variation for the three standard reference materials and the PLCO serum pool combined were calculated using a nested components of variance analysis²⁷ and were 5.9%, 5.1%, and 7.8% respectively for 25(OH)D and 11.5%, 9.4% and 14.9% respectively for DBP. Two controls were missing data on 25(OH)D and one was missing data on DBP; these controls were dropped from the risk analyses.

Statistical analyses

Characteristics of cases and controls were statistically compared using Wilcoxon rank sum and chi-square tests, for continuous and categorical variables, respectively. For continuous variables, correlations among controls were calculated using the Spearman correlation coefficient. A general linear model procedure was used to test the difference of 25(OH)D and DBP concentrations by levels of categorical variables among controls. Latitude was divided into three categories: $< 34^{\circ}\text{N}$, $34\text{--}41^{\circ}\text{N}$ and $> 42^{\circ}\text{N}$. Data suggest that vitamin D cannot be synthesized by skin during the winter months at latitudes above 42°N , but can be synthesized at low concentrations throughout the winter at latitudes below 34°N .²⁸ These categories also correspond to UVB residential regions used in a prior PLCO study of vitamin D and pancreatic cancer, based on Robertson-Berger units.²⁹

Conditional logistic regression was used to determine odds ratios (OR) and 95% confidence intervals (CI) for the association of both vitamin D and DBP with colorectal cancer risk. For 25(OH)D, predefined categories based on clinical definitions in the literature and prior epidemiologic studies^{2,4,18,20,30} were set as < 25 , 25 to < 37.5 , 37.5 to < 50 , 50 to < 75 , 75 to < 100 and ≥ 100 nmol/L, with 50 to < 75 nmol/L chosen as the reference category, because this includes the mean 25(OH)D concentration of the US population.³⁰ Although few subjects had 25(OH)D concentrations ≥ 100 nmol/L, risk for this top category was presented because of the harm noted at higher vitamin D concentrations in a prior cancer study.³¹ Models were conditioned on the matching factors, including date of blood draw. Tests for linear trend were obtained by assigning to each category the median value of controls in that category, and treating this as a continuous variable.

Due to the seasonal variation in 25(OH)D concentrations, three additional approaches were used to examine risk associations with 25(OH)D (although all approaches resulted in similar risk associations). First, season-specific quintiles of 25(OH)D were created based on the

distribution of 25(OH)D among controls in a “darker” (December-May) and a “sunnier” (June-November) season. The season definitions were based on a smoothed plot of weekly predicted 25(OH)D values throughout the year as previously described.²³ These 25(OH)D values peaked in July and were lowest in January. Because the median 25(OH)D concentrations within each of these quintiles differed by season, tests for linear trend were obtained by assigning to each category an ordinal value (1-5) and treating this parameter as a continuous variable. Second, season-adjusted 25(OH)D values were calculated by regressing log-transformed 25(OH)D values against calendar week of blood collection, using a locally weighted polynomial regression method,^{23,32} and creating quintiles of the residuals, based on the control distribution. Finally, the 25(OH)D analyses were stratified on season of blood collection.

DBP quintiles were created based on the distribution among the controls, and entered into the models as indicator variables with the lowest quintile as the referent category. The molar ratio of 25(OH)D:DBP ($\times 10^3$) was used as a proxy for free circulating 25(OH)D.³³ Tests for linear trend were based on the median control values in each category, treated as a continuous variable. Because the monoclonal DBP assay we used may underestimate DBP concentrations in black individuals,³⁴ we also ran all statistical analyses excluding the small number of black subjects from our data; however, as there were no material differences in the results, all data are reported including the black participants.

All variables presented in Table 1 were assessed as potential confounders. Variables which were statistically significantly related to case/control status or to 25(OH)D or DBP were further tested in separate multivariable models for 25(OH)D and DBP. These variables included BMI, physical activity, smoking status, calcium supplement use, and intake of fiber, folate, vitamin D, and calcium (total and dietary intake for the latter three). We also further tested height, vitamin D supplement use (both alone and combined with multivitamin use), total caloric intake, red meat intake, and menopausal hormone therapy (females only). Of these, only BMI was a confounder; none of the other factors changed the ORs by more than 10% in either the 25(OH)D or the DBP models. In addition to models including BMI at study entry, we also present multivariable models adjusted for accepted and putative colorectal cancer risk factors, specifically smoking, physical activity, education, family history of colorectal cancer, alcohol intake, aspirin use, and ibuprofen use. All missing covariate values were retained as a separate category, one subject with missing BMI information was imputed at the median.

Stratified analyses were performed, where possible, using conditional logistic regression, or otherwise using unconditional logistic regression adjusted for the matching factors and BMI. These models were based on quartiles, rather than quintiles, for increased stability. We analyzed 25(OH)D (using season-specific quartiles) and the 25(OH)D:DBP ratio by DBP strata, and we analyzed DBP by 25(OH)D strata. We also examined both 25(OH)D and DBP stratified by age ($\leq 64/ \geq 65$), sex (male/female), race (white/black/other), season of blood collection (December-May/June-November), latitude (defined above), time of blood collection (9 am or earlier/after 9 am), smoking status [never, former (quit ≥ 10 years ago), current & recent (quit < 10 years ago)], BMI (median 26.7 kg/m²), physical activity (< 1 hour of vigorous physical activity per week/ 1 or more hours per week), menopausal

hormone therapy (females only: never, former, current), and follow-up time (median 5.6 years; ≤ 2 years versus > 2 years; and ≤ 5 years versus > 5 years). Models were also examined by cancer stage (I-II versus III-IV), subsite (colon/rectum), and anatomic location within the colon (proximal/distal). Effect modification was statistically evaluated by comparing models with and without a cross-product term of 25(OH)D or DBP (4 categories) and the effect modifier (2-3 categories), using the log-likelihood ratio test. Using continuous versions of the variables, when available, for the cross-product terms did not alter interpretation of the data. To reduce potential influence of subclinical disease on the risk estimates, sensitivity analyses were performed excluding cancers diagnosed during the first two years of follow-up. This also removes cases that were prevalent at baseline or diagnosed because of the baseline screen. Statistical analyses were performed using SAS software version 9.2 (SAS Institute, Inc., Cary, North Carolina) and all P-values were 2-sided.

Results

Median time from blood draw to diagnosis for the cases was 5.6 years (interdecile range 0.3 years to 11.4 years) and median age at cancer diagnosis was 71 years. Compared with controls, there were more former smokers and fewer never smokers among the cases (Table 1). Cases had statistically significantly lower fiber and folate intake and circulating 25(OH)D compared with controls, and a slightly lower 25(OH)D:DBP ratio, but did not differ substantially on other baseline factors.

Among controls, concentrations of 25(OH)D ranged from 28.6-86.5 nmol/L (for 10th-90th percentiles). DBP and 25(OH)D were modestly positively correlated ($r=0.19$, $p<0.0001$). 25(OH)D was inversely correlated with BMI ($r=-0.19$, $p<0.0001$) and positively correlated with intake of dietary vitamin D ($r=0.23$, $p<0.0001$), total vitamin D ($r=0.29$, $p<0.0001$), dietary calcium ($r=0.17$, $p=0.0003$), total calcium ($r=0.20$, $p<0.0001$), dietary folate ($r=0.11$, $p=0.02$), total folate ($r=0.17$, $p=0.0004$) and the 25(OH)D:DBP ratio ($r=0.57$, $p<0.0001$). 25(OH)D concentrations were higher in men than in women (median = 58.9 nmol/L versus 53.9 nmol/L, $p=0.01$), in whites and other races compared with blacks (58.0 nmol/L, 55.7 nmol/L, and 37.4 nmol/L, respectively, $p<0.0001$), for those with blood samples drawn in June-November (61.6 nmol/L versus 49.4 nmol/L for December-May, $p<0.0001$), in vitamin D supplement users (59.3 nmol/L versus 55.2 nmol/L in non-users, $p=0.02$), in those taking supplemental calcium versus not (58.9 nmol/L versus 55.7 nmol/L, $p=0.04$), and in those with one or more hours of vigorous physical activity per week compared with less physical activity (60.3 nmol/L versus 49.7 nmol/L, $p<0.0001$). Concentrations of 25(OH)D did not differ by the other factors in Table 1.

DBP concentrations among controls ranged from 1,946-6,032 nmol/L (for 10th-90th percentiles) and the 25(OH)D:DBP molar ratio ($\times 10^3$) ranged from 7.0-30.9 (for 10th-90th percentiles). Among controls, DBP was inversely correlated with the 25(OH)D:DBP molar ratio ($r=-0.64$, $p<0.0001$). DBP was also modestly inversely correlated with BMI ($r=-0.15$, $p=0.0009$) and DBP concentrations were higher in women than men (median=3,857 nmol/L versus 3,519 nmol/L, $p=0.005$). This difference persisted with further adjustment for 25(OH)D. DBP concentrations did not differ by season, time of blood draw (which ranged from 7 am to 4 pm) or the other factors in Table 1.

Higher concentrations of 25(OH)D were associated with a statistically significantly lower risk of colorectal cancer (Table 2). In analyses using season-specific 25(OH)D quintiles conditioned on the matching factors and adjusted for BMI, colorectal cancer risk for the highest versus lowest quintile was 0.60 (95% CI 0.38-0.94, p-trend 0.01). Excluding cancers diagnosed during the first two years of follow-up did not alter the risk estimates (OR= 0.60, 95% CI 0.35-1.01, p-trend 0.02; based on 365 cases). Further adjustment for recognized colorectal cancer risk factors did not alter the risk estimates; therefore, we subsequently present only the risk estimates conditioned on the matching factors and adjusted for BMI. Using sex- and season-specific quintile cutpoints resulted in little change to the risk estimates (OR=0.64, 95% CI 0.41-0.99 for highest versus lowest quintile, p-trend = 0.03). Risk of colorectal cancer was also lower at the highest 25(OH)D concentration based on a priori cutpoints, although the number of participants in the highest category was small (10 cases/21 controls; OR=0.45, 95% CI 0.20-1.01 for ≥ 100 nmol/L compared with the reference category of 50-< 75 nmol/L, Table 2, and OR=0.31, 95% CI 0.12-0.83 for ≥ 100 nmol/L compared with < 25 nmol/L). Statistically significant inverse associations were also noted when the season-adjusted 25(OH)D values were modeled (OR=0.53, 95% CI 0.33-0.84 for highest versus lowest quintile, p-trend 0.01). The OR for colorectal cancer for a 25 nmol/L increase in serum 25(OH)D was 0.77 (95% CI 0.65-0.91).

DBP was not associated with colorectal cancer risk (Table 2), and incorporating sex-specific cutpoints did not alter the risk estimates (OR=0.79, 95% CI 0.52-1.21, for highest versus lowest quintile, p-trend=0.99). The association was unchanged when cases diagnosed during the first two years of follow-up were excluded (OR=0.73, 95% CI 0.45-1.18, p-trend = 0.79). Adjustment for DBP in any of the 25(OH)D models and vice versa did not affect the risk estimates. The 25(OH)D:DBP molar ratio, a proxy for free 25(OH)D, was not associated with risk in a dose-response manner, although persons in quintiles 3 and 4 appeared to be at lower risk compared with quintile 1.

The lower colorectal cancer risk with higher serum 25(OH)D did not differ by DBP status (Table 3). DBP was not associated with colorectal cancer risk in either the high or low stratum of 25(OH)D, and the 25(OH)D:DBP molar ratio association did not differ substantially by DBP status.

The risk estimates for 25(OH)D did not differ by disease stage or for colon versus rectum (although the number of rectal cancers was relatively small; n=54), but appeared stronger for proximal compared with distal colon cancer (Table 4). Risk also did not differ across subgroups defined by several other characteristics, including age at blood collection, BMI, and time from blood collection to diagnosis (or to equivalent date for controls) (all p-interaction > 0.10, Supplemental Table 1). Some suggestive differences were observed, however. For example, the inverse association was slightly stronger among never-smokers compared with current or former smokers, among participants in the most northern latitudes (> 42° N) versus 34-41° N and < 34° N, in those whose blood was collected December-May compared with June-November, and in those who reported one or more hours of vigorous physical activity per week versus < 1 hour/week. The inverse association for 25(OH)D was also slightly stronger in women than men, and among women who reported current use of menopausal hormone therapy. While the risk estimates for 25(OH)D for whites and blacks

were similar, there was no association for other races, although these results were based on only 41 black cases and 23 of other races.

The DBP association did not differ based on sex, season or time of blood collection, BMI, or other factors tested (all p-interaction >0.16; data not shown). DBP appeared inversely associated with risk among participants with stage I-II disease and for proximal colon cancer, with ORs (95% CI) of 0.65 (0.39-1.06) and 0.68 (0.42-1.09), respectively, but not among individuals with stage III-IV disease or distal colon cancer, with ORs (95% CI) of 1.05 (0.58-1.89) and 1.15 (0.52-2.57) respectively.

A similar percentage of cases and controls reported having a colonoscopy, sigmoidoscopy, or barium enema in the three years prior to randomization in the screening trial (14.3% and 13.5%, respectively). Exclusion of these participants did not alter the risk estimate for 25(OH)D (OR=0.61, 95% CI 0.39-0.96 for highest versus lowest quintile, p-trend=0.03). Likewise, a similar percentage of cases and controls did not complete their expected trial screening (at baseline: 3.4% versus 4.0%, respectively, p=0.61; during follow-up at year three or five: 12.8% versus 10.9%, respectively, p=0.37). Sensitivity analyses excluding subjects unscreened at either baseline or follow-up only slightly attenuated the 25(OH)D association (OR for highest versus lowest quintile = 0.65, 95% 0.41-1.02, p-trend = 0.03).

Discussion

Circulating 25(OH)D was statistically significantly inversely associated with colorectal cancer risk in this large prospective study of men and women, with a 40% lower risk for persons in the highest quintile. The finding was robust across different methods of adjustment for season of blood collection and after controlling for accepted colorectal cancer risk factors, and was similar for colon and rectal cancer. DBP, the major transport protein for 25(OH)D, was not associated with risk and did not modify the 25(OH)D risk association.

Meta-analyses of prospective serologic data indicate an inverse association between circulating 25(OH)D and colorectal cancer risk,¹³⁻¹⁵ with risk estimates of 0.85 (95% CI 0.79-0.91) for a 25 nmol/L increase in 25(OH)D¹³ and 0.66 (0.54-0.81) comparing highest to lowest quantiles.¹⁵ The evidence is far from unequivocal, however. For example, these meta-analyses did not include recent reports from the Alpha-Tocopherol, Beta-Carotene Cancer Prevention (ATBC) Study, which found statistically significantly higher risk in men with higher serum 25(OH)D,²⁰ or the ESTHER study in Germany, which was null,³⁵ and only one included the null findings from the Physicians' Health Study.¹⁵ In addition, only four individual studies reported statistically significantly lower risk estimates or trends for colorectal cancer.¹⁶⁻¹⁹ Thus, the current data, from the second largest investigation to date, substantially strengthen the evidence that higher vitamin D status is associated with a lower risk of colorectal cancer.

The inverse 25(OH)D association was somewhat stronger in women (OR=0.48) than in men (OR=0.66), although the interaction term was not statistically significant. Data are limited in this regard, with one study reporting nonsignificantly lower risk of rectal cancer in women and an opposite association in men,²² another suggesting a positive colon cancer association

in women but not in men,²¹ and the large European Prospective Investigation into Cancer and Nutrition (EPIC) study showing a similar association in women and men.¹⁸ Whether the modest sex difference we observed is related to our finding of an inverse 25(OH)D association that was noticeably stronger among women receiving menopausal hormone therapy at baseline than among never users (OR = 0.33 and 0.65, respectively) requires further study, but implicates a role for sex hormones in the vitamin D-colorectal cancer association.

Our examination of the vitamin D-colorectal cancer association found a similar inverse association in blacks and whites. The Multiethnic Cohort Study did not find differences across Latino, African-American, Hawaiian, Japanese-American, and white populations,¹⁷ but an inverse association was reported in a Japanese population.²¹ Additional studies in diverse populations are clearly needed.

Prospective serologic data are inconsistent regarding the vitamin D association for colon compared with rectal cancer.¹⁷⁻²² In addition, statistically significant elevated risk with higher vitamin D concentrations was noted for colon cancer in ATBC²⁰ and positive (albeit, nonsignificant) associations were also noted for colon cancer in a Japanese study²¹ and for rectal cancer in the Health Professionals Follow-up Study.²² In the present investigation, risks for colon and rectal cancer were ~40% and 50% lower, respectively, in the highest vitamin D quartile, although the latter estimate was based on relatively few cases. We observed an inverse association for proximal, but not distal, colon cancer; the lack of association for the latter could be explained by the PLCO sigmoidoscopy screening, which reduced the number of distal colon cancers, and which may have resulted in unusual distal cancers (*e.g.*, an overrepresentation of aggressive cancers).

Studies examining genetic variants in the vitamin D pathway with risk of colorectal cancer, offering an alternative approach to testing the vitamin D hypotheses, have focused primarily on the vitamin D receptor (*VDR*) gene, but have been inconclusive.¹⁴ Evidence for other variants, such as in *GC* (the gene encoding DBP), *CYP27B1*, and *CYP24A1* and colorectal cancer is also inconclusive,³⁶ with a large consortium analysis finding no association between SNPs in the four key vitamin D genes (*GC*, *CYP2R1*, *CYP24A1* and *DHCR7/NADSYN1*) and risk of colorectal cancer.³⁷

Controlled trials of supplemental vitamin D provide little evidence for a reduction in colorectal cancer incidence.^{16,38-40} The earliest trial tested 100,000 IU of vitamin D every four months (~833 IU daily) against placebo for five years in 2,686 participants and reported no effect on colon cancer (HR = 1.02, 95% CI 0.60-1.74) based on 55 incident cases.³⁸ The Women's Health Initiative (WHI), by far the largest trial with over 36,000 participants, provided 400 IU vitamin D daily (along with 1g calcium) or placebo for an average of seven years and reported a hazard ratio of 1.08 (95% CI: 0.86-1.34) for colorectal cancer incidence.¹⁶ Trial participants were permitted to self-supplement with vitamin D up to 600 IU/day, however, and a secondary analysis showed a nonsignificant reduction in risk (HR=0.81, 95% CI 0.58-1.13 for the calcium plus vitamin D arm) among the 15,302 women who did not self-supplement.⁴¹ A smaller trial that used 1,100 IU vitamin D combined with 1.4-1.5g calcium daily for four years in 1,179 women reported a statistically significant

reduction in total cancer incidence but had only five colon cancers.³⁹ Explanations for the divergent findings of the controlled trials and prospective studies include short durations of the trials (especially since cancer has a long latency), contamination of the placebo group by participants who self-supplemented (e.g., as in WHI), inclusion of participants in trials who are already replete for vitamin D, insufficient power for colorectal cancer in the trials, and potential selection bias and uncontrolled confounding in observational studies.^{3,42}

Various anti-carcinogenic mechanisms for vitamin D have been proposed, including reduction of cancer cell proliferation, angiogenesis, and inflammation, and stimulation of differentiation and apoptosis.¹⁻³ These are in large part mediated through VDR: 1,25(OH)₂D binds to VDR, which in turn creates a heterodimer with the retinoid X receptor (RXR), and binds specific DNA response elements to activate or inhibit target genes. Over 1,600 binding sites for VDR/RXR have been identified in a colorectal cancer cell line, including for the genes *c-FOS* and *c-MYC*, which are both associated with cell growth control. A direct effect of cellular 1,25(OH)₂D on these genes would be consistent with a potential anti-carcinogenic mechanism for vitamin D in the colon.⁴³ Circulating 1,25(OH)₂D is tightly regulated, but the discovery that VDR and the 1 α -hydroxylase enzyme that converts 25(OH)D to 1,25(OH)₂D, are expressed in many extra-renal tissues including the colon, suggests that 1,25(OH)₂D could inhibit carcinogenesis locally in the colon.^{2,3} Another proposed mechanism is up-regulation of detoxification of harmful secondary bile acids in the colon by vitamin D-activated VDR.⁴⁴

DBP binds and transports vitamin D and its metabolites with high affinity^{1,5} such that only 0.04% of 25(OH)D and 0.4% of 1,25(OH)₂D are unbound (i.e., in a “free” state).⁵ Less than 5% of DBP is bound to vitamin D or its metabolites because DBP circulates at much higher concentrations (μ M) than vitamin D (nM).⁶ Circulating DBP is reported to be relatively stable during an individual’s lifespan, but may be influenced by pregnancy or estrogen therapy,⁶ as well as by SNPs in the *GC* gene.⁴⁵ DBP displays diurnal variation with a nadir at 4 am⁴⁶ but unlike 25(OH)D, DBP does not vary seasonally.^{6,33,46} In our analysis, neither time of day of blood collection nor use of menopausal hormone therapy was associated with DBP concentrations or modified the associations, although, similar to the sex difference reported in another study,⁴⁷ we observed median circulating DBP to be 10% higher in women than men.

The prospective design of the PLCO study, with blood collected from all participants at baseline and up to 18 years of follow-up, are strengths, as they minimize the potential for serum vitamin D to be altered by colorectal cancer or by cancer treatment. A unique aspect of our study is that both cases and controls underwent sigmoidoscopic screening at baseline and three or five years later, with similar adherence rates, thereby reducing the potential confounding influence of screening behavior. We measured 25(OH)D, the accepted biomarker of vitamin D status, and DBP, the major vitamin D transport protein, and the inverse 25(OH)D-colorectal cancer association was evident using several methods to account for seasonal variation. While we cannot completely rule out residual confounding as an explanation of our findings, we adjusted our models for multiple recognized potential confounding factors with no noticeable effect on risk estimates. Vitamin D and DBP were measured only once, and while blood sampling at additional time-points might give a more

accurate estimate of usual vitamin D status, methodological studies have shown 25(OH)D concentrations to be well-correlated over a 3-14 year time period (correlations of 0.50-0.70),⁴⁸⁻⁵⁰ and 25(OH)D remained stable in blood collected 20 years apart and stored for 10, 20, or 30 years.⁵¹ DBP concentrations appear to be stable over the lifetime⁶ and were correlated in samples collected up to three years apart (adjusted ICC = 0.97).⁵² Although our study included a large number of men and women with colorectal cancer (second in size only to the EPIC study¹⁸), the number of rectal cancer cases was small. Similar to most studies, we had limited power to examine non-whites and specific subgroups such as smokers.

In conclusion, higher 25(OH)D was associated with statistically significantly lower colorectal cancer risk in this large prospective study in the U.S. We observed modest risk differences by sex and no material differences between white and black participants or for colon versus rectal cancers. DBP was not associated with risk, and we found no evidence of effect modification of the 25(OH)D association by DBP, or vice versa. A large international pooling project of prospective studies of breast and colorectal cancers in progress, which will take advantage of a wide range of circulating vitamin D concentrations, should be poised to confirm the 25(OH)D association reported here and examine subgroup differences with greater power.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

Acknowledgments

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Abbreviations

1,25(OH)₂D	1,25-dihydroxyvitamin D
25(OH)D	25-hydroxyvitamin D
ATBC	Alpha-Tocopherol, Beta-Carotene Cancer Prevention Study
CI	confidence interval
DBP	vitamin D binding protein
EPIC	European Prospective Investigation into Cancer and Nutrition Study
ICD	International Classification of Diseases
OR	odds ratio

PLCO	Prostate, Lung, Colorectal, and Ovarian Cancer Screening Trial
VDR	Vitamin D Receptor
WHI	Women's Health Initiative

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Novelty and impact statement

Prospective studies of circulating 25-hydroxyvitamin D, the accepted biomarker of vitamin D status, suggest an inverse association with colorectal cancer risk, but with some inconsistencies and no consideration of the key transport protein, vitamin D binding protein (DBP). The current study, which eliminates confounding by colorectal cancer screening behavior, finds that higher vitamin D status is associated with substantially lower colorectal cancer risk, but does not support a direct or modifying role for DBP.

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Table 1Selected baseline characteristics for cases and controls¹

Characteristic	Cases, n=476	Controls, n=476	p ²
Age at study entry (years)	64 (61-68)	64 (61-68)	3
Females	206 (43.3%)	206 (43.3%)	3
Race			3
White	412 (86.6%)	412 (86.6%)	
Black	41 (8.6%)	41 (8.6%)	
Other	23 (4.8%)	23 (4.8%)	
Blood collected December-May	229 (48.1%)	227 (47.7%)	3
Body mass index (kg/m ²)	27.2 (24.5-30.5)	26.7 (24.5-29.9)	0.39
Height (cm)	172.7 (162.6-177.8)	170.2 (162.6-177.8)	0.81
Vigorous physical activity			0.79
< 1 hours/week	137 (28.8%)	141 (29.6%)	
1-3 hours/week	203 (42.7%)	206 (43.3%)	
≥ 4 hours/week	105 (22.1%)	96 (20.2%)	
Missing	31 (6.5%)	33 (6.9%)	
College graduate	147 (30.9%)	150 (31.5%)	0.83
Family history of colorectal cancer (% yes)	62 (13.0%)	46 (9.7%)	0.17
Pre-trial colorectal cancer screening (% yes) ⁴	68 (14.3%)	64 (13.5%)	0.80
Smoking history			0.07
Never-smoker	190 (39.9%)	218 (45.8%)	
Former smoker (quit ≥ 10 years ago)	168 (35.3%)	161 (33.8%)	
Former smoker (quit < 10 years ago)	43 (9.0%)	25 (5.3%)	
Current smoker	52 (10.9%)	44 (9.2%)	
Pipe/cigar only	18 (3.8%)	26 (5.5%)	
Missing	5 (1.1%)	2 (0.4%)	
Aspirin use (% yes)	222 (47%)	240 (50.4%)	0.30
Ibuprofen use (% yes)	137 (28.8%)	126 (26.5%)	0.61
Supplement use ⁵			
Calcium (% yes)	203 (42.7%)	206 (43.3%)	0.98
Vitamin D (% yes)	186 (39.1%)	190 (39.9%)	0.96
Dietary intake/day			
Energy (kcal)	1,903 (1,473-2,552)	1,955 (1,520-2,538)	0.72
Fiber (g)	21.3 (15.5-28.9)	22.9 (16.9-29.8)	0.04
Vitamin D (IU)	186 (123-266)	187 (130-279)	0.43
Vitamin D including supplements (IU)	334 (167-594)	341 (175-583)	0.94
Calcium (mg)	850(605-1,165)	884 (627-1,194)	0.25
Calcium including supplements (mg)	1,029 (731-1,528)	1,096 (778-1,615)	0.23
Folate (ug)	340 (250-453)	371 (266-484)	0.04
Folate including supplements (ug)	502 (304-738)	498 (328-755)	0.61
Alcohol (g)	1.46 (0.21-12.89)	1.02 (0.21-9.10)	0.24

Characteristic	Cases, n=476	Controls, n=476	p ²
Serum biomarkers			
25(OH)D (nmol/L)	52.3 (39.6-64.8)	56.5 (41.2-71.1)	0.01
DBP (nmol/L)	3,775 (2,579-4,890)	3,584 (2,715-4,938)	0.96
25(OH)D:DBP ratio ($\times 10^3$)	14.1 (9.7-20.6)	15.4 (10.4-21.5)	0.07

¹ Values are medians (interquartile range) or number (percent); Abbreviations: 25(OH)D, 25-hydroxyvitamin D; DBP, vitamin D binding protein

² P-values are based on Wilcoxon rank sum test (for continuous variables) and chi-square test (for categorical variables), all p-values are 2-sided

³ Matching factor

⁴ A "yes" means the subject had a colonoscopy, sigmoidoscopy, or barium enema, during the 3 years prior to joining the Trial

⁵ Current supplement use includes individual supplements and the contribution from multivitamins

Table 2Associations between serum 25(OH)D, DBP and the 25(OH)D:DBP molar ratio and risk of colorectal cancer¹

	Cases/Controls, N	BMI-adjusted OR (95% CI) ²	Multivariate- adjusted OR (95% CI) ³
25(OH)D - season-specific quintiles ⁴			
Quintile 1	109/96	1.00 (reference)	1.00 (reference)
Quintile 2	119/95	1.07 (0.73-1.56)	1.05 (0.70-1.16)
Quintile 3	97/96	0.84 (0.56-1.25)	0.89 (0.58-1.36)
Quintile 4	82/93	0.76 (0.50-1.14)	0.80 (0.52-1.25)
Quintile 5	69/94	0.60 (0.38-0.94)	0.59 (0.36-0.95)
p-trend ⁵		0.01	0.02
25(OH)D - a priori defined cutpoints, nmol/L			
<25.0	39/33	1.42 (0.81-2.51)	1.26 (0.69-2.30)
25.0 - <37.5	64/62	1.10 (0.73-1.66)	1.19 (0.78-1.83)
37.5 - <50.0	116/93	1.37 (0.95-1.98)	1.32 (0.90-1.94)
50.0 - <75.0	186/193	1.00 (reference)	1.00 (reference)
75.0 - < 100	61/72	0.86 (0.58-1.28)	0.87 (0.58-1.33)
≥100	10/21	0.45 (0.20-1.01)	0.40 (0.17-0.92)
p-trend ⁶		0.01	0.02
DBP, nmol/L			
Quintile 1: ≤ 2,460	111/95	1.00 (reference)	1.00 (reference)
Quintile 2: > 2,460 and ≤ 3,188	73/95	0.65 (0.43-1.00)	0.63 (0.40-0.98)
Quintile 3: > 3,188 and ≤ 4,087	89/95	0.77 (0.51-1.18)	0.76 (0.49-1.18)
Quintile 4: > 4,087 and ≤ 5,215	111/95	1.00 (0.66-1.51)	1.01 (0.65-1.56)
Quintile 5: > 5,215	92/95	0.82 (0.54-1.26)	0.80 (0.51-1.26)
p-trend ⁶		0.95	0.94
25(OH)D:DBP molar ratio (×10 ³) ⁷			
Quintile 1: ≤ 9.49	115/95	1.00 (reference)	1.00 (reference)
Quintile 2: > 9.49 and ≤ 13.64	113/95	0.96 (0.64-1.44)	0.99 (0.64-1.53)
Quintile 3: > 13.64 and ≤ 17.21	79/95	0.66 (0.43-1.01)	0.67 (0.42-1.04)
Quintile 4: > 17.21 and ≤ 23.29	74/95	0.61 (0.39-0.94)	0.61 (0.38-0.96)
Quintile 5: > 23.29	95/94	0.79 (0.52-1.21)	0.82 (0.52-1.28)
p-trend ⁶		0.18	0.24

¹ Abbreviations: 25(OH)D, 25-hydroxyvitamin D; CI, confidence interval; DBP, vitamin D binding protein; OR, odds ratio² Values are odds ratios and 95% confidence intervals, calculated from logistic regression models, conditioned on age at study entry (+/- 1 year), sex, race (white, black, other), and date of serum collection (+/- 30 days, but expanded to +/- 60 days for 2 subjects), and adjusted for body mass index (<25, 25-30, ≥30 kg/m²).³ Values calculated as above, with additional adjustment for smoking (never, former quit ≥10 years ago, former quit < 10 years ago, current, pipe/cigar, missing), physical activity (< 1 hours/week, 1-3 hours/week, ≥ 4 hours/week, missing), education (less than high school, completed high school, vocational school, some college, college graduate, graduate studies, missing), family history of colorectal cancer (yes, no, missing), alcohol intake (<1 g/day, ≥1-15 g/day, ≥15-30 g/day, >30 g/day, missing), aspirin use (yes, no, missing), and ibuprofen use (yes, no, missing).

⁴ Cutpoints for the season-specific quintiles for December-May were Q1: ≤ 33.6 , Q2: > 33.6 and ≤ 44.6 , Q3: > 44.6 and ≤ 55.7 , Q4: > 55.7 and ≤ 68.0 , Q5: > 68.0 nmol/L; and for June-November were Q1: ≤ 44.1 , Q2: > 44.1 and ≤ 57.3 , Q3: > 57.3 and ≤ 66.4 , Q4: > 66.4 and ≤ 78.2 , Q5: > 78.2 nmol/L.

⁵ Tests for linear trend were obtained by assigning to each category an ordinal value (1-5) and treating this as a continuous variable. All P-values are 2-sided.

⁶ Tests for linear trend were obtained by assigning to each category its median and treating this as a continuous variable. All P-values are 2-sided.

⁷ A proxy for free 25(OH)D

Association between serum 25(OH)D and the 25(OH)D:DBP molar ratio stratified by DBP, and between serum DBP stratified by 25(OH)D, and risk of colorectal cancer¹

	Quartile 1	Quartile 2	Quartile 3	Quartile 4	p-trend ²	p-interaction
25(OH)D, season-specific ³						
Cases/controls, N	84/71	55/64	54/60	29/42		
OR (95% CI) ⁴	1.00 (reference)	0.74 (0.45-1.23)	0.78 (0.47-1.30)	0.59 (0.32-1.08)	0.11	0.60
		DBP below median				
Cases/controls, N	60/50	73/52	65/60	56/75		
OR (95% CI) ⁴	1.00 (reference)	1.10 (0.64-1.87)	0.82 (0.48-1.40)	0.57 (0.33-0.97)	0.02	
		DBP above median				
Range (nmol/L)	≤ 2715	>2715 and ≤ 3584	>3584 and ≤ 4938	>4938		
		25(OH)D below median ⁵				
Cases/controls, N	86/73	52/62	74/47	58/55		
OR (95% CI) ⁴	1.00 (reference)	0.69 (0.42-1.14)	1.28 (0.77-2.11)	0.86 (0.52-1.15)	0.85	0.52
		25(OH)D above median				
Cases/controls, N	52/46	32/56	65/72	57/63		
OR (95% CI) ⁴	1.00 (reference)	0.51 (0.28-0.93)	0.81 (0.47-1.38)	0.81 (0.47-1.42)	0.92	
25(OH)D:DBP molar ratio (×10 ³) ⁶						
Range	≤ 10.4	>10.4 and ≤ 15.4	>15.4 and ≤ 21.5	>21.5		
		DBP below median				
Cases/controls, N	26/22	41/42	52/63	103/110		
OR (95% CI) ⁴	1.00 (reference)	0.85 (0.42- 1.75)	0.70 (0.35- 1.39)	0.82 (0.43-1.56)	0.64	0.87
		DBP above median				
Cases/controls, N	114/97	92/76	41/56	7/8		
OR (95% CI) ⁴	1.00 (reference)	0.96 (0.63- 1.46)	0.58 (0.35-0.96)	0.72 (0.25- 2.10)	0.06	

¹ Abbreviations: 25(OH)D, 25-hydroxyvitamin D; CI, confidence interval; DBP, vitamin D binding protein; OR, odds ratio

- ²Tests for linear trend for season-specific 25(OH)D were obtained by assigning to each category an ordinal value (1-4) and for DBP and 25(OH)D:DBP by assigning the median of each category, and treating each as a continuous variable. All p-values are 2-sided.
- ³Cutpoints for the season-specific quartiles for December-May were Q1: ≤ 37.2 , Q2: > 37.2 and ≤ 49.4 , Q3: > 49.4 and ≤ 64.5 , Q4: > 64.5 nmol/L; cutpoints for the season-specific quartiles for June-November were Q1: ≤ 47.3 , Q2: > 47.3 and ≤ 61.6 , Q3: > 61.6 and ≤ 75.6 , Q4: > 75.6 nmol/L.
- ⁴Values are odds ratios and 95% confidence intervals, calculated from unconditional logistic regression models, adjusted for age at study entry (< 59 , 60-64, 65-69, > 70), sex (male/female), race (white, black, other), date of blood collection (continuous), and body mass index (< 25 , 25- < 30 , ≥ 30 kg/m²)
- ⁵Based on separate medians calculated for each season
- ⁶A proxy for free 25(OH)D

Association between season-specific serum 25(OH)D and risk of colorectal cancer stratified on disease stage, subsite, and anatomic location¹

Table 4

Stage	Quartile 1 ²	Quartile 2	Quartile 3	Quartile 4	p-trend ³
Stage I-II					
Cases/controls, N	84/65	64/61	69/77	56/68	
OR (95% CI) ⁴	1.00 (reference)	0.80 (0.49-1.31)	0.68 (0.42-1.10)	0.62 (0.37-1.03)	0.05
Stage III-IV					
Cases/controls, N	59/54	63/54	49/43	28/48	
OR (95% CI) ⁴	1.00 (reference)	1.04 (0.61-1.78)	0.99 (0.56-1.78)	0.51 (0.27-0.95)	0.05
Subsite					
Colon					
Cases/controls, N	126/109	116/103	107/108	72/99	
OR (95% CI) ⁴	1.00 (reference)	0.96 (0.66-1.40)	0.84 (0.57-1.23)	0.61 (0.40-0.93)	0.02
Rectum					
Cases/controls, N	18/12	12/13	12/12	13/18	
OR (95% CI) ⁴	1.00 (reference)	0.61 (0.20-1.85)	0.68 (0.22-2.14)	0.52 (0.17-1.59)	0.30
Anatomic location					
Proximal colon					
Cases/controls, N	91/76	82/76	79/72	48/75	
OR (95% CI) ⁴	1.00 (reference)	0.87 (0.55-1.36)	0.87 (0.54-1.37)	0.49 (0.30-0.82)	0.01
Distal colon					
Cases/controls, N	35/33	34/26	28/35	22/24	
OR (95% CI) ⁴	1.00 (reference)	1.27 (0.61-2.63)	0.76 (0.37-1.58)	0.96 (0.42-2.18)	0.57
Distal colon & rectum					
Cases/controls, N	53/45	46/39	40/47	35/42	
OR (95% CI) ⁴	1.00 (reference)	1.01 (0.56-1.84)	0.73 (0.40-1.33)	0.74 (0.39-1.41)	0.23

¹ Abbreviations: 25(OH)D, 25-hydroxyvitamin D; CI, confidence interval; OR, odds ratio

² Cutpoints for the season-specific quartiles for December-May were Q1: ≤ 37.2, Q2: > 37.2 and ≤ 49.4, Q3: > 49.4 and ≤ 64.5, Q4: > 64.5 nmol/L; cutpoints for the season-specific quartiles for June-November were Q1: ≤ 47.3, Q2: > 47.3 and ≤ 61.6, Q3: > 61.6 and ≤ 75.6, Q4: > 75.6 nmol/L

³ Tests for linear trend were obtained by assigning to each category an ordinal value (1-4) and treating this as a continuous variable. All P-values are 2-sided.

⁴ Odds ratios and 95% confidence intervals were calculated using unconditional regression, adjusted for age at study entry (< 59, 60-64, 65-69, > 70), sex (male/female), race (white, black, other), date of blood collection (continuous), and body mass index (<25, 25-<30, ≥30 kg/m²)